

Preguntas:

1. Sistema operativo y versión.

R: Windows 11

2. Distribución (solamente en el caso de Linux).

R:

Distributor ID: Debian

Description: Debian GNU/Linux 12 (bookworm)

Release: 12

Codename: bookworm

3. Versión de la instalación.

R:

Docker: Docker version 29.7.2

PostgreSQL: postgres (PostgreSQL) 18.6

DBeaver: DBeaver 26.2.0

4. Tiempo requerido.

R: Aproximadamente 40 minutos.

5. Explicación del paso a paso que realizaste con sus respectivas capturas de pantallas.

(Adicionalmente agrega las evidencias de los pasos que consideres esenciales de la instalación).

6. Comentarios y los problemas a los que te enfrentaste en la instalación.

R: En el paso 6 de "Instalación Docker" hubo un error de creación de repositorio. Se tomo otro comando que prácticamente contenía el código provisto por el profesor, pero en una sola línea.

Instalación Docker:

Dado que mi Sistema operativo es Windows 11 se realizó la siguiente tarea sobre un subsistema llamado WSL.

1. Para instalar el repositorio oficial, primero actualizaremos la lista de paquetes existentes:

```

PS C:\Users\sooye> wsl
sooye(main) ► sudo apt update
[sudo] password for sooye:
Hit:1 http://deb.debian.org/debian bookworm InRelease
Get:2 http://security.debian.org/debian-security bookworm-security InRelease [34.8 kB]
Get:3 https://dl.google.com/linux/chrome-stable/deb stable InRelease [2,548 B]
Get:4 http://deb.debian.org/debian bookworm-updates InRelease [55.4 kB]
Get:5 http://ftp.debian.org/debian bookworm-backports InRelease [59.4 kB]
Get:6 http://security.debian.org/debian-security bookworm-security/main amd64 Packages [335 kB]
Get:7 http://security.debian.org/debian-security bookworm-security/main Translation-en [209 kB]
Get:8 https://dl.google.com/linux/chrome-stable/deb stable/main amd64 Packages [1,410 B]
Fetched 698 kB in 1s (1.043 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
80 packages can be upgraded. Run 'apt list --upgradable' to see them.

```

2. Instalamos los paquetes necesarios para agregar el repositorio de Docker:

```

sooye(main) ► sudo apt install ca-certificates curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libcurl3-gnutls libcurl4
The following packages will be upgraded:
  ca-certificates curl libcurl3-gnutls libcurl4
4 upgraded, 0 newly installed, 0 to remove and 56 not upgraded.
Need to get 1,256 kB of archives.
After this operation, 3,072 B of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bookworm/main amd64 curl amd64 7.88.1-10+deb12u15 [316 kB]
Get:2 http://deb.debian.org/debian bookworm/main amd64 libcurl4 amd64 7.88.1-10+deb12u15 [392 kB]
Get:3 http://deb.debian.org/debian bookworm/main amd64 libcurl3-gnutls amd64 7.88.1-10+deb12u15 [386 kB]
Get:4 http://security.debian.org/debian-security bookworm-security/main amd64 ca-certificates all 20250419-deb12u1 [162 kB]
Fetched 1,256 kB in 0s (4,393 kB/s)
Preconfiguring packages ...
(Reading database ... 136775 files and directories currently installed.)
Preparing to unpack .../ca-certificates_20250419-deb12u1_all.deb ...
Unpacking ca-certificates (20250419-deb12u1) over (20230311-deb12u1) ...
Preparing to unpack .../curl_7.88.1-10+deb12u15-amd64.deb ...
Unpacking curl (7.88.1-10+deb12u15) over (7.88.1-10+deb12u14) ...
Preparing to unpack .../libcurl4_7.88.1-10+deb12u15-amd64.deb ...
Unpacking libcurl4:amd64 (7.88.1-10+deb12u15) over (7.88.1-10+deb12u14) ...
Preparing to unpack .../libcurl3-gnutls_7.88.1-10+deb12u15-amd64.deb ...
Unpacking libcurl3-gnutls:amd64 (7.88.1-10+deb12u15) over (7.88.1-10+deb12u14) ...
Setting up libcurl3-gnutls:amd64 (7.88.1-10+deb12u15) ...
Setting up ca-certificates (20250419-deb12u1) ...
Updating certificates in /etc/ssl/certs...
rehash: warning: skipping ca-certificates.crt, it does not contain exactly one certificate or CRL
21 added, 13 removed; done.
Setting up libcurl4:amd64 (7.88.1-10+deb12u15) ...
Setting up curl (7.88.1-10+deb12u15) ...

Processing triggers for man-db (2.11.2-2) ...
Processing triggers for libc-bin (2.36-9+deb12u14) ...
Processing triggers for ca-certificates (20250419-deb12u1) ...
Updating certificates in /etc/ssl/certs...
0 added, 0 removed; done.
Running hooks in /etc/ca-certificates/update.d...
done.
Processing triggers for ca-certificates-java (20230710-deb12u1) ...
done.
Scanning processes...

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

3. Creamos el directorio donde se almacenaran las llaves de los repositorios:

```

sooye(main) ► sudo install -m 0755 -d /etc/apt/keyrings

```

4. Descargamos la llave GPG oficial de Docker:

```

sooye(main) ► sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc

```

5. Damos permisos de lectura:

```

sooye(main) ► sudo chmod a+r /etc/apt/keyrings/docker.asc

```

6. Agregamos el repositorio oficial de Docker:

```

sooye(main) ► sudo tee /etc/apt/source.list.d/
docker.sources <<EOF
Types: deb
URIs:
https://download.docker.com/linux/ubuntu
Suites: $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}")
Components: stable
Architectures: $(dpkg
--print-architecture)
Signed-By: /etc/apt/keyrings/
docker.asc
EOF
tee: /etc/apt/source.list.d/: Is a directory

```

Dado que no se creo correctamente el repositorio se ejecuto el siguiente comando, misma pero de una sola línea:

```
sooye(main) ▶ echo "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keysrings/docker.asc] https://download.docker.com/linux/$(. /etc/os-release && echo "$ID") $(. /etc/os-release && echo "$VERSION_CODENAME") stable" | sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
sooye(main) ▶ sudo apt update
Get:1 https://download.docker.com/linux/debian bookworm InRelease [46.6 kB]
Hit:2 http://ftp.debian.org/debian bookworm-backports InRelease
Get:3 https://download.docker.com/linux/debian bookworm/stable amd64 Packages [79.2 kB]
Hit:4 http://deb.debian.org/debian bookworm InRelease
Hit:5 http://security.debian.org/debian-security bookworm-security InRelease
Hit:6 http://deb.debian.org/debian bookworm-updates InRelease
Hit:7 https://dl.google.com/linux/chrome-stable/deb stable InRelease
Fetched 126 kB in 1s (225 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
56 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

7. Actualizamos el paquete de base de datos con los paquetes de Docker

```
sooye(main) ▶ sudo apt update
Hit:1 http://deb.debian.org/debian bookworm InRelease
Hit:2 http://deb.debian.org/debian bookworm-updates InRelease
Hit:3 http://security.debian.org/debian-security bookworm-security InRelease
Hit:4 https://dl.google.com/linux/chrome-stable/deb stable InRelease
Hit:5 http://ftp.debian.org/debian bookworm-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
56 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

8. Instalamos Docker

```
sooye(main) ▶ sudo apt install docker-ce docker-ce-cli containerd.io
docker-buildx-plugin docker-compose-plugin
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  criu libintl-perl libintl-xs-perl libmodule-find-perl libnet1 libnl-3-200 libproc-processtable-perl libprotobuf-c1 libprotobuf32
  libsort-naturally-perl libterm-readkey-perl needrestart python3-protobuf tini
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  docker-buildx-plugin docker-ce-rootless-extras docker-compose-plugin pigz
Suggested packages:
  docker-model-plugin
The following packages will be REMOVED:
  containerd docker.io runc
The following NEW packages will be installed:
  containerd.io docker-buildx-plugin docker-ce docker-ce-cli docker-ce-rootless-extras docker-compose-plugin pigz
0 upgraded, 7 newly installed, 3 to remove and 56 not upgraded.
Need to get 102 MB of archives.
After this operation, 134 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://deb.debian.org/debian bookworm/main amd64 pigz amd64 2.6-1 [64.0 kB]
Get:2 https://download.docker.com/linux/debian bookworm/stable amd64 containerd.io amd64 2.3.4-1-debian.12-bookworm [22.7 MB]
Get:3 https://download.docker.com/linux/debian bookworm/stable amd64 docker-ce-cli amd64 5:29.7.2-1-debian.12-bookworm [17.0 MB]
Get:4 https://download.docker.com/linux/debian bookworm/stable amd64 docker-ce amd64 5:29.7.2-1-debian.12-bookworm [24.0 MB]
Get:5 https://download.docker.com/linux/debian bookworm/stable amd64 docker-buildx-plugin amd64 0.36.1-1-debian.12-bookworm [17.2 MB]
Get:6 https://download.docker.com/linux/debian bookworm/stable amd64 docker-ce-rootless-extras amd64 5:29.7.2-1-debian.12-bookworm [10.2 MB]
Get:7 https://download.docker.com/linux/debian bookworm/stable amd64 docker-compose-plugin amd64 5.5.0-1-debian.12-bookworm [11.1 MB]
Fetched 102 MB in 6s (15.8 MB/s)

(Reading database ... 136783 files and directories currently installed.)
Removing docker.io (20.10.24+dfsg1-1+deb12u1+b6) ...
'/usr/share/docker.io/contrib/nuke-graph-directory.sh' -> '/var/lib/docker/nuke-graph-directory.sh'
Removing containerd (1.6.20-dsl-1+deb12u3) ...
Removing runc (1.1.5+ds1-1+deb12u1) ...
Selecting previously unselected package containerd.io.
(Reading database ... 136504 files and directories currently installed.)
Preparing to unpack .../8-containerd.io_2.3.4-1-debian.12-bookworm_amd64.deb ...
Unpacking containerd.io (2.3.4-1-debian.12-bookworm) ...
Selecting previously unselected package docker-ce-cli.
Preparing to unpack .../1-docker-ce-cli_5%3a29.7.2-1-debian.12-bookworm_amd64.deb ...
Unpacking docker-ce-cli (5:29.7.2-1-debian.12-bookworm) ...
Selecting previously unselected package docker-ce.
Preparing to unpack .../2-docker-ce_5%3a29.7.2-1-debian.12-bookworm_amd64.deb ...
Unpacking docker-ce (5:29.7.2-1-debian.12-bookworm) ...
Selecting previously unselected package pigz.
Preparing to unpack .../3-pigz_2.6-1_amd64.deb ...
Unpacking pigz (2.6-1) ...
Selecting previously unselected package docker-buildx-plugin.
Preparing to unpack .../4-docker-buildx-plugin_0.36.1-1-debian.12-bookworm_amd64.deb ...
Unpacking docker-buildx-plugin (0.36.1-1-debian.12-bookworm) ...
Selecting previously unselected package docker-ce-rootless-extras.
Preparing to unpack .../5-docker-ce-rootless-extras_5%3a29.7.2-1-debian.12-bookworm_amd64.deb ...
Unpacking docker-ce-rootless-extras (5:29.7.2-1-debian.12-bookworm) ...
Selecting previously unselected package docker-compose-plugin.
Preparing to unpack .../6-docker-compose-plugin_5.5.0-1-debian.12-bookworm_amd64.deb ...
Unpacking docker-compose-plugin (5.5.0-1-debian.12-bookworm) ...
Setting up docker-buildx-plugin (0.36.1-1-debian.12-bookworm) ...
Setting up containerd.io (2.3.4-1-debian.12-bookworm) ...
Installing new version of config file /etc/containerd/config.toml ...
Setting up docker-compose-plugin (5.5.0-1-debian.12-bookworm) ...
Setting up docker-ce-cli (5:29.7.2-1-debian.12-bookworm) ...
Setting up pigz (2.6-1) ...
Setting up docker-ce-rootless-extras (5:29.7.2-1-debian.12-bookworm) ...
Setting up docker-ce (5:29.7.2-1-debian.12-bookworm) ...
```

```
Installing new version of config file /etc/default/docker ...
Installing new version of config file /etc/init.d/docker ...
Processing triggers for man-db (2.11.2-2) ...
Scanning processes...

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
zsh: command not found: docker-buildx-plugin
```

9. Comprobamos el servicio:

```
sooye(main) ► sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-09-01 18:51:45 CST; 1min 41s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 8613 (dockerd)
      Tasks: 14
     Memory: 27.7M
        CPU: 466ms
    CGroup: /system.slice/docker.service
           └─8613 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 01 18:51:44 sooyeon dockerd[8613]: time="2026-09-01T18:51:44.519293547-06:00" level=info msg="Deleting nftables IPv4 rules" error="running"
Sep 01 18:51:44 sooyeon dockerd[8613]: time="2026-09-01T18:51:44.539823988-06:00" level=info msg="Deleting nftables IPv6 rules" error="running"
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.332257399-06:00" level=info msg="Loading containers: done."
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.377328826-06:00" level=info msg="Docker daemon" commit=6a43e3d containerd-snap
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.377418823-06:00" level=info msg="Initializing buildkit"
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.378554832-06:00" level=warning msg="failed check for feaverity support" error="
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.504518785-06:00" level=info msg="Completed buildkit initialization"
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.515754118-06:00" level=info msg="Daemon has completed initialization"
Sep 01 18:51:45 sooyeon dockerd[8613]: time="2026-09-01T18:51:45.515822927-06:00" level=info msg="API listen on /run/docker.sock"
Sep 01 18:51:45 sooyeon systemd[1]: Started docker.service - Docker Application Container Engine.
```

10. Probamos si podemos trabajar con imágenes de Docker:

```
sooye(main) ► sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:5dd0d36e255913fc30f90b9f2b1d359cc2cbdb4808cc4b65f1676e203243cc
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/
```

SMBD PostgreSQL:

1. Necesitamos descargar la imagen de postgres, el cual Docker utilizara al momento de crear el contenedor.

```
sooye(main) ► docker pull postgres
Using default tag: latest
latest: Pulling from library/postgres
6310eb16bf42: Pull complete
6463c6b5c703: Pull complete
8d6422da287d: Pull complete
2fc0e69f8d7d: Pull complete
5fa29c524b4e: Pull complete
ace4b5b96a91: Pull complete
af2a6480a058: Pull complete
5ee6c6ca7fd9: Pull complete
c5694bfff1357: Pull complete
aedab7177504: Pull complete
a4635eae9b8: Pull complete
46e663abc8f5: Pull complete
1bb54e8f7727: Pull complete
Digest: sha256:4ef4dbc939d61acea57712655ddb4b4ab27419c913f94cca0cd57cb3ea3c2280
Status: Downloaded newer image for postgres:latest
docker.io/library/postgres:latest
```

2. Una vez descargado la imagen de postgres, utilizaremos el siguiente comando para crear el contenedor.

```
sooye(main) ▶ docker run -d \
--name postgres \
-e POSTGRES_PASSWORD= \
-p 127.0.0.1:5432:5432 \
-v postgres_data:/var/lib/postgresql \
postgres
8dd559206648662903c49ec0267a73fe19845d5abe1949381a97668100df95ce
```

- Una vez creado el contenedor, automáticamente se ejecutara, pero si necesitas inicializarlo una vez ya creado y que no esté activo tenemos que averiguar el CONTAINER ID de PostgreSQL, con el siguiente comando:

```
sooye(main) ▶ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
8dd559206648   postgres  "docker-entrypoint.s..." About a minute ago Up About a minute    127.0.0.1:5432->5432/tcp    postgres
e5d3fe00c12c   hello-world  "/hello"                9 minutes ago Exited (0) 9 minutes ago                               vigilant_shaw
```

- Supongamos que por el anterior comando nos da que el CONTAINER ID es a1e321c26d33, entonces para iniciar el contenedor utilizamos:

```
sooye(main) ▶ docker start e5d3fe00c12c
e5d3fe00c12c
```

- Y para detener el contenedor que estamos utilizando, utilizamos el comando siguiente.

```
sooye(main) ▶ docker stop e5d3fe00c12c
e5d3fe00c12c
```

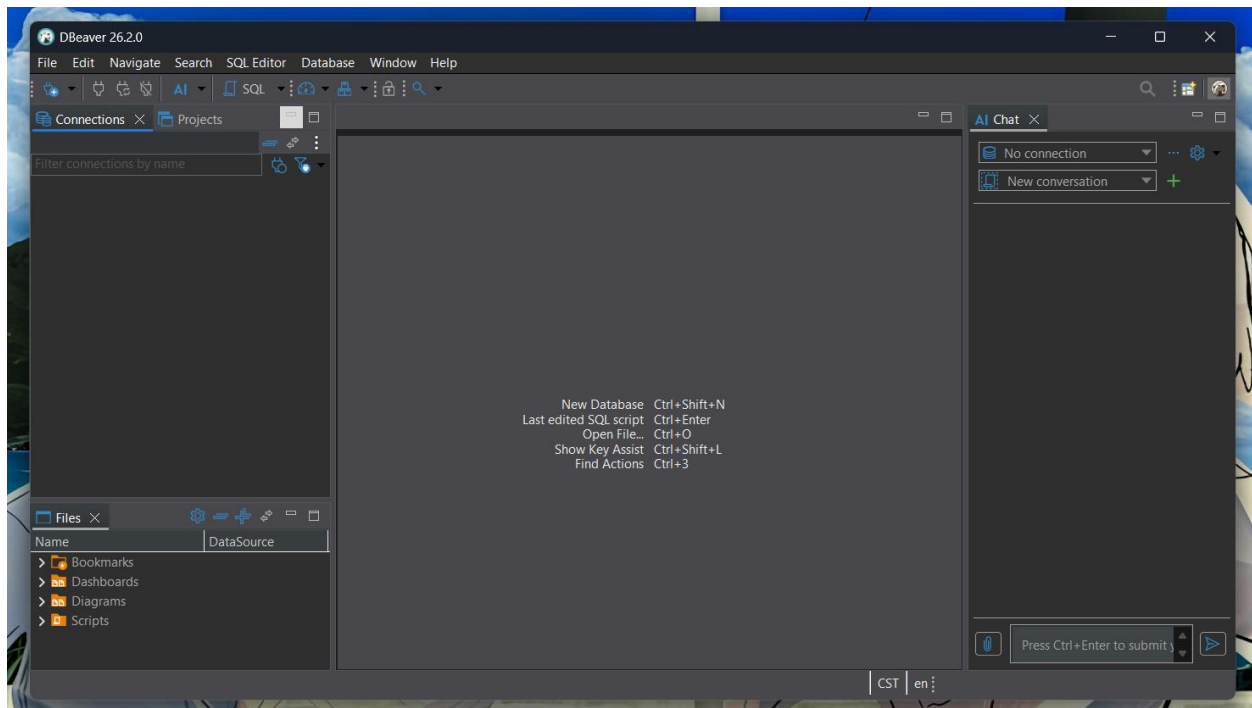
- Si queremos conectarnos a PostgreSQL una vez creado el contenedor podemos utilizar psql(La consola de comandos de PostgreSQL) con:

```
sooye(main) ▶ docker run -it --link postgres:postgres postgres psql -h postgres -U postgres
WARNING: Links on the default bridge network are deprecated and will be removed in a future release. Use a custom network instead.
Password for user postgres:
psql (18.6 (Debian 18.6-1.pgdg13+2))
Type "help" for help.

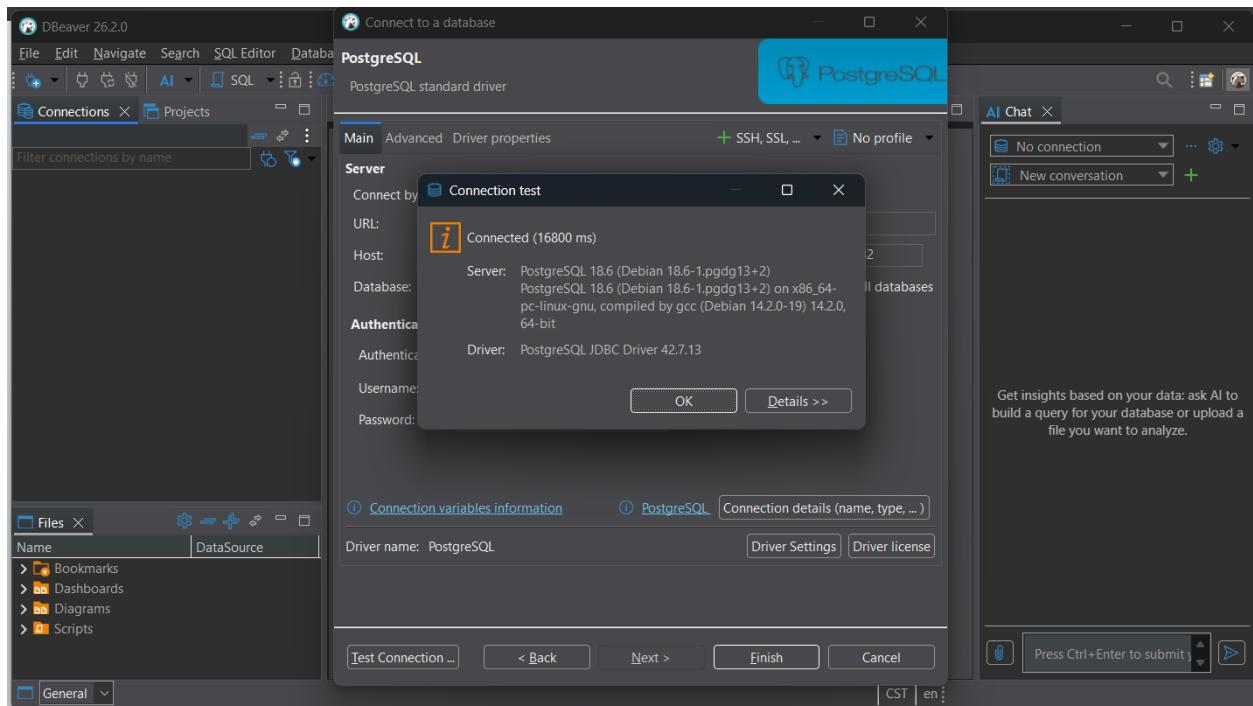
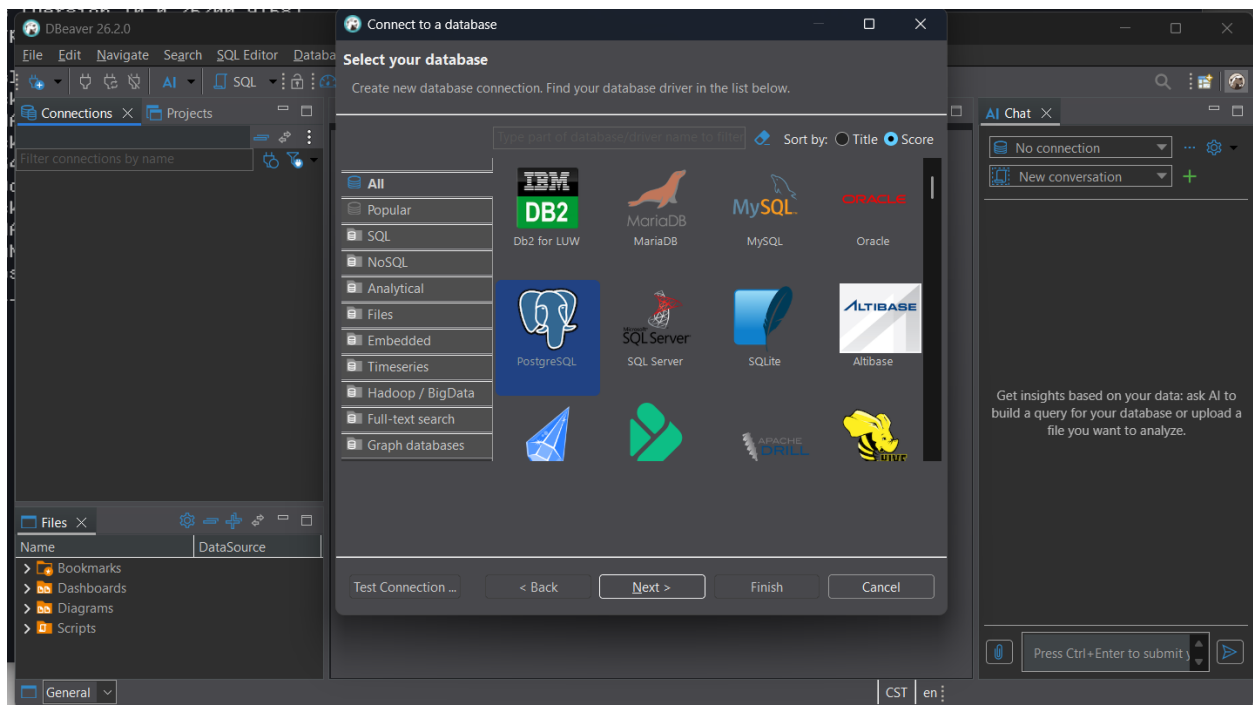
postgres=# q
postgres=# \q
```

Database Tools:

Se instalo DBeaver



Conexión:



Verificación:

DBeaver 26.2.0 - <postgres> Script

File Edit Navigate Search SQL Editor Database Window Help

Commit Rollback Auto postgres public@postgres

Connections Projects

Filter connections by name

postgres localhost:5432

select version()

Results 1

select version() Enter a SQL expression to filter results (use ▶)

Grid	Value
1	PostgreSQL 18.6 (Debian 18.6-1.pgdg13+2) on x86_64-pc

Refresh Save Cancel Export data

200 1 1 row(s) fetched - 0.004s, on 2026-09-05 at 10:56:06

General

CS en Writable Smart Insert 1..16

AI Chat

postgres

New conversation

Get insights based on your data: ask AI to build a query for your database or upload a file you want to analyze.

Press Ctrl+Enter to submit